

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

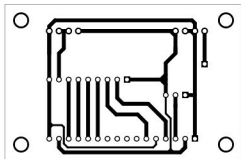


Fig. 5: An actual-size PCB pattern of the receiver circuit

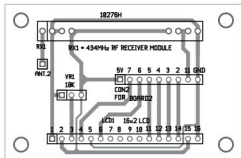


Fig. 6: Component layout of the receiver PCB

The receiver part contains ASK RF receiver module RX1, which re-

ceives the data serially sent from TX1. This serial data is given to digital pin 11 of Arduino2. A 16x2 LCD is interfaced with Arduino2, which displays the data.

Construction and testing

An actual-size, single-side PCB of the transmitter is shown in Fig. 3 and its component layout in Fig. 4. Similarly, an actual-size, single-side PCB layout of the receiver is shown in Fig. 5 and its component layout in Fig. 6.

Now, it is time to test the project. First, install VirtualWire.h library in Arduino IDE before uploading the programs (sketches) to respective Arduino boards. Then, upload the transmitter sketch in Arduino1. Next, upload the receiver sketch to Arduino2, which has the LCD connected to it. (Receiver sketch requires VirtualWire and

LiquidCrystal libraries.)

Now, connect Arduino1 to a laptop/PC. Open serial monitor in Arduino IDE using Ctrl+Shift+M. Check if baud rate is 9600; if not, change it to 9600. Enter a string or text such as Hello Arduino. Remember to terminate it with a full stop. Press Enter. The message will be displayed on the LCD at the receiving Arduino2. To get proper display on the LCD, adjust the contrast control 10-kilo-ohm preset (VR1). **EFY**



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